
ASSIGNMENT 1

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Question-3

One Step of Gradient Descent in a Deep Neural Network (25 marks)

Consider the following feedforward neural network with three affine layers (four layers including the input), ReLU activations in the hidden layers, and two output classes.

Network architecture

- **Input:**

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$$

- **Layer 1 (Affine + ReLU):**

$$z^{(1)} = W^{(1)}x + b^{(1)}, \quad a^{(1)} = \text{ReLU}(z^{(1)})$$

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Layer 2 (Affine + ReLU):**

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}, \quad a^{(2)} = \text{ReLU}(z^{(2)})$$

$$W^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Layer 3 (Output layer, affine only):**

$$s = W^{(3)}a^{(2)} + b^{(3)}$$

$$W^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(3)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

There are two classes, and the true class label is $y = 1$ (using 1-based indexing). The learning rate is $\eta = 0.1$.

Assume the derivative of the ReLU function is

$$\text{ReLU}'(t) = \begin{cases} 1 & t > 0 \\ 0 & t \leq 0 \end{cases}$$

Part A: Softmax with Cross-Entropy Loss**(12 Marks)**

The softmax function is defined as

$$q_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$$

and the cross-entropy loss for a single example is

$$L_{CE} = -\log q_y$$

- (a) Perform the forward pass and compute $z^{(1)}, a^{(1)}, z^{(2)}, a^{(2)}, s$, and the softmax probabilities q . (3)
- (b) Compute the cross-entropy loss L_{CE} . (1)
- (c) Compute the gradient of the loss with respect to the logits s . (1)
- (d) Using backpropagation, compute the gradients with respect to all weights and biases:

$$\nabla W^{(3)}, \quad \nabla b^{(3)}, \quad \nabla W^{(2)}, \quad \nabla b^{(2)}, \quad \nabla W^{(1)}, \quad \nabla b^{(1)} \quad (4)$$

- (e) Perform one gradient descent update step for all parameters and report the updated values. (3)

Part B: Multiclass Hinge Loss**(13 marks)**

The multiclass hinge loss is defined as

$$L_{\text{hinge}} = \sum_{j \neq y} \max(0, s_j - s_y + 1)$$

- (a) Compute the hinge loss for the given example. (2)
- (b) Compute the gradient of the hinge loss with respect to the logits s . (2)
- (c) Using backpropagation, compute the gradients with respect to all weights and biases. (6)
- (d) Perform one gradient descent update step for all parameters and report the updated values. (3)